MIL-STD-2156(AS)
4 APRIL 1983

# MILITARY STANDARD

# LAUNCHER, RAIL, GUIDED MISSILE, AIRCRAFT, GENERAL DESIGN CRITERIA FOR

FSC 1440

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DEPARTMENT OF DEFENSE
Washington, DC 20301

Launcher, Rail, Guided Missile, Aircraft, General Design Criteria for

MIL-STD-2156(AS)

1. This Military Standard is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.



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# 1. SCOPE

1.1 Scope. The purpose of this document is to provide guidelines for establishing general criteria for the design, development, construction, maintenance and testing of a rail launcher. This standard applies to all rail launchers including those which mount directly to the aircraft structure and those supported by a 14-inch or 30-inch suspension system per MIL-A-8591.



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## 2. REFERENCED DOCUMENTS

2.1 Issues of Documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this standard to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| L-P-383 | Plastic Material, Polyester Resin, Glass Fiber Base, Low Pressure Laminated. |
| QQ-A-367 | Aluminum Alloy Forgings. |
| QQ-P-416 | Plating, Cadmium (electrodeposited). |

MILITARY

| | |
|---|---|
| DoD-D-1000 | Drawings, Engineering and Associated Lists. |
| MIL-S-5002 | Surface Treatments and Metallic Coatings for Metal Surfaces of Weapons Systems. |
| MIL-B-5087 | Bonding, Electrical, and Lightning Protection, for Aerospace Systems. |
| MIL-W-5088 | Wiring, Aircraft, Selection and Installation of. |
| MIL-C-5541 | Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys. |
| MIL-H-5606 | Hydraulic Fluid, Petroleum Base, Aircraft, Missile and Ordnance. |
| MIL-C-6021 | Casting, Classification and Inspection of. |
| MIL-E-6051 | Electromagnetic Compatibility Requirements, Systems. |
| MIL-H-6088 | Heat Treatment, Aluminum Alloys. |
| MIL-H-6875 | Heat Treatment of Steels (Aircraft Practice) Process for. |
| MIL-F-7179 | Finishes and Coatings, General Specification for Protection of Aerospace Weapons, Structures and Parts. |
| MIL-F-7190 | Forgings, Steel, for Aircraft and Special Ordnance Applications. |



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SPECIFICATIONS - Continued.

MILITARY

| | |
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| MIL-T-7743 | Testing, Store Suspension and Release Equipment, General Specification for. |
| MIL-B-7883 | Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys. |
| MIL-I-8500 | Interchangeability and Replaceability of Component Parts for Aerospace Vehicles. |
| MIL-A-8591 | Airborne Stores, Associated Suspension Lugs, and Aircraft-Store Interface (Carriage Phase); General Design Criteria for. |
| MIL-A-8625 | Anodic Coatings, for Aluminum and Aluminum Alloys. |
| MIL-A-8837 | Coating, Cadmium (Vacuum Deposited). |
| MIL-L-8937 | Lubricant, Solid Film, Heat Cured. |
| MIL-P-15024 | Plates, Tags and Bands for Identification of Equipment. |
| MIL-F-18264 | Finishes, Organic, Weapons System, Application and Control of. |
| MIL-N-18307 | Nomenclature and Identification. |
| MIL-A-21180 | Aluminum Alloy Castings, High Strength. |
| MIL-A-22771 | Aluminum Alloy Forgings, Heat Treated. |
| MIL-C-38999 | Connectors, Electrical, Circular, Miniature High Density, Quick Disconnect, Environment Resistant, Removable Crimp Contacts. |
| MIL-H-46855 | Human Engineering Requirements for Military Systems, Equipment and Facilities. |
| MIL-H-81200 | Heat Treatment of Titanium and Titanium Alloys. |
| MIL-F-83142 | Forging, Titanium Alloys, for Aircraft and Aerospace Applications. |
| MIL-C-85485 | Cable, Electric, Filter Line, Radio Frequency Absorptive. |



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STANDARDS

MILITARY

| | |
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| MIL-STD-130 | Identification Marking of U.S. Military Property. |
| MIL-STD-143 | Specifications and Standards, Order of Precedence for the Selection of. |
| MIL-STD-454 | Standard General Requirements for Electronic Equipment. |
| MIL-STD-461 | Electromagnetic Compatibility. |
| MIL-STD-462 | Electromagnetic Interference Characteristics, Measurement of. |
| MIL-STD-810 | Environmental Test Methods. |
| MIL-STD-838 | Lubrication of Military Equipment. |
| MIL-STD-882 | System Safety Program Requirements. |
| MIL-STD-889 | Dissimilar Metals. |
| MIL-STD-1385 | Preclusion of Ordnance Hazards. |
| MIL-STD-1472 | Human Engineering Design Criteria for Military Systems Equipment and Facilities. |
| MIL-STD-1568 | Materials and Processes for Corrosion Prevention and Control in Aerospace Weapons Systems. |
| MIL-STD-1587 | Materials and Processes Requirements for Air Force Weapons Systems. |
| MIL-STD-1760 | Aircraft/Store Electrical Interconnection System. |
| MIL-STD-2088 | Bomb Rack Unit (BRU), Aircraft, General Design Criteria For. |

HANDBOOKS

MILITARY

| | |
|---|---|
| MIL-HDBK-5 | Metallic Materials and Elements for Aerospace Vehicle Structures. |
| MIL-HDBK-235 | Electromagnetic (Radiated) Environment Considerations for Design and Procurement of Electrical and Electronic Equipment, Subsystems and Systems-Part 1A. |



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PUBLICATIONS

MILITARY

| | |
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| MIL-BUL-147 | Specifications and Standards of Non-Government Organizations Released for Flight Vehicle Construction. |

NAVAL AIR SYSTEMS COMMAND

| | |
|---|---|
| AR-43 | Electromagnetic Compatibility Advisory Board; Requirement for. |
| SD-24 | General Specification for Design and Construction of Aircraft Weapons Systems. |
| WR-62 | Naval Weapons Requirements, Specifications and Standards; Use of. |
| 19-15BD-6 | Technical Manual, "Description, Operation and Maintenance Instruction with Illustrated Parts Breakdown for Single Hoist Ordnance Loading System (SHOLS)." |
| 19-100-2 | Airborne Weapons Handling Equipment (Shipboard). |
| 19-100-1.1 (Vol 1)<br>19-100-1.2 (Vol 2) | Approved Handling Equipment for Weapons and Explosives, Volumes 1 and 2. |

CHIEF OF NAVAL OPERATIONS

| | |
|---|---|
| OPNAVINST 4790.2 | Naval Aviation Maintenance Program. |

AIR STANDARDIZATION COORDINATING COMMITTEE - AIR STANDARDS

| | |
|---|---|
| AIR STD 20/16 | Design Guide to Preclude Hazards of Electromagnetic Radiation to Airborne Weapon Systems. |
| AIR STD 20/18 | Laboratory Tests for Stores Suspension Equipment. |

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)



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## 3. DEFINITIONS

3.1 Missile launcher. An item rigidly attached to an aircraft to carry, service, launch and jettison air-launched missiles.

3.2 Air-launched missile. A guided, self-propelled store designed to be launched from an airborne vehicle and whose target is either airborne, on the ground or under the water surface.

3.3 Rail launcher. A launcher containing rails on which the missile is carried, and along which the missile travels after initiation of the missile's self-propulsion system.

3.4 Ejection launcher. A launcher which provides an initial source of energy to adequately displace the missile from the aircraft prior to the initiation of the missile's self-propulsion system.

3.5 Launch. The intentional separation of the missile from the aircraft for normal employment of the missile.

3.6 Jettison. The intentional separation of the launcher or the launcher/missile from the aircraft in a safed/unarmed condition.

3.7 Arming. The process of removing the safety devices and closing the signal paths necessary to allow firing of the missile motor or detonation of the missile warhead.

3.8 Safe separation. The parting of a missile from an aircraft without damage to, contact with or adverse effects on the aircraft, its launchers and other weapons.

3.9 Acceptable separation. Acceptable store separations satisfy not only safe separation criteria but also pertinent operational criteria to meet guidance control and trajectory requirements.

3.10 Hung Missile. A missile which does not separate from the launcher when the launch cycle is initiated.

3.11 Loading. The operation of installing missiles on the launcher and aircraft.

3.12 Suspension system. Launcher elements which engage the missile to react missile triaxial forces and moments, and generally consist of a launcher rail, support structure, snubbing provisions and longitudinal restraint device.

3.13 Missile snubbing device. A device for maintaining rigidity between the missile and the launcher rail.

3.14 Release system. The release system normally consists of the missile propulsion system and a launcher detent mechanism.

3.15 Detent mechanism. A launcher mechanism for providing longitudinal restraint to the missile.



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3.16 Locking system. An electromechanical device for insuring locking of the detent mechanism.

3.17 Umbilical. Normally a coolant line or an electrical harness which attaches to the missile with a quick disconnect fitting or plug. It is used to control or test missile equipment while the missile is attached to the launcher and aircraft.

3.18 Pylon. A pylon is a suspension device externally attachable on the wing or fuselage of an aircraft, with provisions for attaching aircraft stores.



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# 4. GENERAL REQUIREMENTS

4.1 Launcher types. Rail launchers are generally used for missiles weighing less than 350 pounds, and can be used for missiles weighing up to 800 pounds. Ejection launchers can also be used for missiles weighing 350 to 800 pounds. Missiles weighing more than 800 pounds are assumed to be carried only on ejection launchers.

4.2 Selection of materials, specifications, standards and drawings. The selection of materials, standard parts, processes, corrosion protection, and design features significant in corrosion behavior shall be in accordance with the requirements of Design Specification SD-24, MIL-STD-1568 and MIL-STD-1587.

4.2.1 Materials. Materials shall conform to applicable specifications, be compatible with environmental and service conditions (see 4.3.10) and shall be as specified herein and on applicable drawings. Design shall make maximum use of standard (MS, AN, MIL-STD, etc) parts, materials and processes, rather than special or peculiar items. Materials which are not covered by government specifications or which are not specifically described herein, shall be of the best quality, suitable for the purpose intended. Particular care shall be given to close fitting parts in the choice of both materials and corrosion prevention method. Materials shall be selected such that wet lubricants or preservatives are not required, and the use of dry lubricants is minimized.

4.2.1.1 Metal parts. All metal parts shall be of the corrosion resistant type or treated in a manner to render them resistant to corrosion. Type AISI 431 corrosion resistant steel shall not be used. Unless suitably protected against electrolytic corrosion, dissimilar metals, as defined in MIL-STD-889, shall not be used in contact with each other. General design information governing usage of metals is furnished in MIL-HDBK-5.

4.2.1.1.1 Heat treatment. Heat treatment of aluminum, steel and titanium parts shall be in accordance with MIL-H-6088, MIL-H-6875, and MIL-H-81200 respectively.

4.2.1.1.2 Castings. Castings used in the launcher shall conform with the requirements of MIL-C-6021 with appropriate class, grade and critical area notations. In addition, aluminum alloy castings, used in critical strength applications, shall conform to the requirements of MIL-A-21180.

4.2.1.1.3 Forgings. Forgings used in the launcher shall conform to the requirements of MIL-F-7190, MIL-F-83142, or QQ-A-367 with appropriate grade and grain flow notations. Forgings used in critical strength applications shall conform to the requirements of MIL-F-7190 Grade A or MIL-A-22771.

4.2.1.2 Non-metallic components. Non-metallic components shall be designed for minimum deterioration caused by abrasion, exposure to sunlight, microorganisms, moisture, temperature extremes, fuel hydraulic and lubricating oil, grease, and salt spray. Protection shall be provided for those non-metallic components for which strength degradation associated with abrasion, load or exposure-induced deterioration can endanger or jeopardize the function of the launcher.



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4.2.1.2.1 Reinforced plastic construction. Reinforced plastic materials if required, shall be as specified for Type I materials in L-P-383 and shall be of such character and quality as to be capable of withstanding all service conditions, as herein specified, without degrading the performance of the component of the launcher.

4.2.1.3 Lubrication. Lubricants and lubrication practices shall conform to the requirements of MIL-STD-838 and MIL-L-8937 as applicable. Lubricants shall function satisfactorily throughout the temperature range from minus 70°F to plus 280°F. Choice of lubricants shall (a) be compatible with non-metallic components, (b) not damage finishes adjacent to location of lubricant application, (c) eliminate the need for frequent lubrication by field maintenance activities and (d) be non-reactive with the environment. If lubrication is required, choice of lubricants and practices shall be such that lubrication need be accomplished only during post deployment intermediate level maintenance (see 4.3.11.2.2).

4.2.1.4 Hydraulic fluids. Hydraulic fluids used in the launcher should be in accordance with MIL-H-5606.

4.2.1.5 Fungus-proof materials. To the greatest extent practicable, the materials used in the launcher shall be non-nutrients for fungi.

4.2.1.6 Potting compounds. Potting compounds employed in the launcher shall comply with requirement 47 of MIL-STD-454.

4.2.1.7 Corrosion protection. Corrosion protective practices employed in the manufacture of the launcher shall be in accordance with the MIL-F-7179 requirements for exterior surfaces. Design of the launcher shall make use of materials which preclude corrosion susceptibility under service environmental conditions without a requirement for hermetic sealing.

4.2.1.8 Finishes. Protective coatings and finishes shall not crack, chip, or scale during normal service, or in the herein specified extremes of environmental conditions. Surface treatments, coatings and finishes shall conform to MIL-S-5002, or surface treatments specified herein. General guidance in the application and control of organic finishes is provided in MIL-F-18264.

4.2.1.8.1 Anodizing. All non-fatigue critical aluminum and aluminum alloy parts, not subject to wear, shall have Type II anodic coatings in accordance with MIL-A-8625. Aluminum and aluminum alloy parts subject to wear shall have Type III anodic coatings in accordance with MIL-A-8625.

4.2.1.8.2 Chemical surface treatment. For aluminum and aluminum alloy parts not subject to wear, abrasion or erosion, chemical conversion surface treatment in accordance with MIL-C-5541 may be used in lieu of anodizing.

4.2.1.8.3 Plating. Plating shall be avoided where possible. When required, plating of steel surfaces shall be in accordance with the requirements of MIL-S-5002. Steel parts not subject to wear, in contact with aluminum or aluminum alloys, shall be cadmium plated in accordance with QQ-P-416, Type II, Class I, or MIL-A-8837, Type II, Class 1.



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4.2.1.9 Electrical connectors. Connectors shall be "scoop proof" to prevent accidental shorting of or damage to contacts during mating of the plug and receptacle. The connectors shall contain positive mating features to indicate completed connection and to prevent any possibility of mismating. Connectors shall contain peripheral grounding fingers that connect the two mating halves of the connector before pin and socket contacts make connection. The connector shall also contain provisions for terminating shield braid for electromagnetic interference (EMI) protection. The aircraft, weapon and launcher sides of the electrical connection shall contain socket-type contacts, and the electrical harness shall contain pin-type contacts. If there is no electrical harness, then the aircraft or launcher side of the electrical connection shall contain socket-type contacts, and the weapon side shall contain pin-type contacts. Electrical connector selection shall be made utilizing MIL-C-38999, to the series identified in MIL-STD-1760. Soldered or brazed electrical connections shall not be used in electrical connectors.

4.2.1.10 Wiring. All electrical wiring shall be in accordance with MIL-W-5088 and MIL-C-85485.

4.2.1.11 Soldered or brazed connections. The soldering of contacts shall be in accordance with MIL-STD-454 requirement 5. When a brazing process is used, it shall be in accordance with MIL-B-7883.

4.2.2 Specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143 and WR 62. A partial listing of approved MIL-STD-143 Group II non-government organization specifications and standards is furnished in MIL-BUL-147.

4.2.3 Drawings. Drawing requirements shall be specified by the procuring activity in accordance with DoD-D-1000 instructions.

4.3 General design requirements.

4.3.1 Performance. The launcher shall provide for safe and reliable captive carriage, service and launch of a specified missile configuration installed on tactical aircraft. The launcher shall be capable of processing aircraft power and aircraft/missile signals essential to missile system operation and test. Launcher design shall permit quick and reliable loading and downloading of the missile to the launcher, and of the launcher to the aircraft, and provide easy access for maintenance and servicing. The launcher shall sustain specified design loads and environments which shall not adversely affect the launcher operation.

4.3.2 Design factors of safety. The launcher yield factor of safety shall be 1.15 times the design limit load. Design limit loads are the maximum actual loads that may be expected to occur in service. No permanent deformation shall be allowed after application of design yield loads. The launcher ultimate factor of safety shall be 1.5 times the design limit load. Failure of the launcher upon application of the ultimate load shall be defined as:

a. Separation of the missile from the launcher, or failure of the launcher to retain its missile in normal carriage position.

b. Separation of the launcher from the aircraft.

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Other structural breakage or permanent deformation of the launcher does not constitute failure at the ultimate load level. In addition, all castings used shall have a minimum margin of safety of 0.33, all fittings/forgings shall have a minimum margin of safety of 0.20. Allowable yield and ultimate loads shall include the effects of thermal environments.

4.3.2.1 Pressure systems. Components of the pressure system shall be designed to a proof pressure of 2.0 and burst pressure of 2.5 times the operational peak pressure of that system.

4.3.3 Structural loads requirements. Load analyses and tests shall be conducted to determine launcher design loads for the following conditions.

4.3.3.1 Take-off and landing loads. The launcher shall withstand the maximum loads imposed by catapult take-off and arrested landing in accordance with MIL-A-8591. Applicable inertial acceleration limits and loads shall be developed.

4.3.3.2 Captive flight loads. The launcher shall be designed for the maximum aerodynamic and inertial loads acting on the missile and launcher throughout the captive flight mach-altitude envelope of the aircraft (minimum store configuration). The design captive flight envelope and the inertial acceleration levels of the most severe aircraft/weapon station shall be used to develop worst-case launcher design loads in accordance with MIL-A-8591.

4.3.3.3 Launch loads. The launcher shall be designed for the maximum aerodynamic and inertial loads acting on the missile and launcher throughout the specified launch mach-altitude envelope of the aircraft (minimum store configuration). Combined launcher loads shall include launch forces and the structural dynamic interactions which occur at missile separation.

4.3.3.4 Jettison loads. The launcher shall be designed to withstand the combined effect of maximum ejection forces imposed by the supporting bomb rack and by the aerodynamic and inertial forces acting on the missile/launcher combination over the specified jettison envelope.

4.3.3.5 Cyclic loads. The launcher shall be capable of withstanding the cyclic application of combined limit loads imposed upon it over the course of its specified life. The cyclic load spectra shall include the effects of specified environments and a minimum of 500 launches.

4.3.4 Stress analysis. A stress analysis shall be accomplished using the loads generated in accordance with 4.3.2 and 4.3.3. Margins of safety shall be verified by static tests and cyclic load tests of 4.3.3.5 as required by the procuring activity.

4.3.5 Service life. The launcher design objective shall be to meet or exceed an economically maintainable service life of 10 years exclusive of shelf life based on anticipated aircraft mission profiles. An economically maintainable service life is exceeded when the cost to repair the launcher is more than one-half the replacement cost. In terms of missile release, the launcher shall have a minimum design service life of 500 missile launches prior to major overhaul.



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4.3.6 Reliability. Mission reliability for a launcher shall be a minimum of 0.99 at 90 percent confidence level. Mission reliability is defined as the probability that the launcher shall successfully launch missiles within the specified mission parameters. The shelf life of the launcher and all components shall be a minimum of five years, and shall be refurbishable following the minimum shelf life specified. A maximum number of parts (subject to approval by the procuring agency) shall have a shelf life which is at least ten times the service life of the launcher, and a service life which is greater than that of the launcher.

4.3.7 Interfaces.

4.3.7.1 Launcher to aircraft interface. The launcher to aircraft interface shall be a function of the ejection rack used to support the rail launcher. In some cases, such as a wingtip launcher installation, the launcher shall be a fixed non-jettisonable installation. In either case, the launcher mounting shall be such as to minimize degradation of aircraft/missile performance, weight and drag, and shall utilize optimum load paths. Launcher design shall be tailored to existing aircraft support structure, shall not require special adaptors, and shall provide ease of installation and removal. When a launcher is required to satisfy different aircraft weapon stations which differ in primary support provisions, missile orientation or clearance requirements, the use of special adaptors may be permitted to avoid major aircraft modifications and to enhance system versatility. Physical interfaces between the launcher and aircraft shall be defined in terms of dimensions and tolerances and shall address each applicable aircraft weapon station. Functional interfaces shall include the support, service, missile control and fire control system functions. Electrical interfaces shall include aircraft supplied power and logic and control signals to and from the captive missile. Appropriate grounding and shielding shall be provided. All interface requirements shall comply with applicable aircraft/missile interface specifications. During jettisoning, proper electrical connector separation must be accomplished at the ejection rack to launcher interface.

4.3.7.2 Launcher to missile interface. Interfaces between the launcher and the missile shall include missile support, snubbing and retention, and the pre-launch, launch, and service functions. Location of arming units, suspension and snubbing provisions shall be such as to optimize missile carry and separation performance and minimize missile stresses and weight. Physical interfaces including service umbilicals and safe-arm mechanisms shall be specified in terms of dimensions and tolerances under both static and dynamic conditions. Electrical interfaces shall specify missile power, and logic and control signal requirements. Appropriate grounding and shielding shall be provided. All interface requirements shall comply with applicable aircraft/missile interface specifications.

4.3.7.3 Launcher to support equipment interface. The functional and physical interfaces between the launcher and all applicable support equipment (SE) shall be specified and shall include any government furnished equipment (GFE) or contractor furnished equipment (CFE) used for handling, captive support, transporting, aircraft loading, cleaning, ground testing and servicing, and shipboard storage of the launcher.



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4.3.7.4 Electrical grounding interface. The grounding system design shall provide compatibility with the grounding system within the aircraft and all other equipment to be used with or which interfaces with the launcher. The grounding system shall be in accordance with 5.12.5.

4.3.7.4.1 Launcher to aircraft structure grounding. The launcher structure shall be electrically grounded to the aircraft through the physical, mechanical interface of the two structures. The preparation of the conducting path and choice of mating surfaces composition, material and electrical characteristics shall be submitted by the contractor to the procuring activity for approval.

4.3.7.4.2 Launcher to internal ground interface. Safety grounds shall be provided integral with the harness and connectors. The launcher shall be grounded in a manner which will prevent ground loops and ground returns common to signal and power circuits and will provide effective shielding for signal circuits, minimize electromagnetic interference, and protect personnel from electrical hazards.

4.3.8 Physical characteristics. The physical characteristics of the launcher shall be such as to satisfy all functional requirements. Major components of the launcher shall include as a minimum basic structure, retention mechanism, support and snubbing assemblies, latching and locking mechanisms, umbilical mechanisms, harnesses and rocket motor firing circuits.

4.3.8.1 Launcher weight. The launcher shall be designed for minimum weight consistent with functional requirements, physical interface requirements and ease of handling during installation and maintenance, to the limits defined in the applicable detailed specification.

4.3.8.2 Launcher drag. The launcher shall be designed for minimum drag consistent with functional requirements, physical interface requirements and ease of handling during installation and maintenance, to the limits defined in the applicable detailed specification.

4.3.8.3 Launcher radar cross section. The launcher shall be designed for minimum radar cross section consistent with functional requirements, physical interface requirements and ease of handling during installation and maintenance, to the limits defined in the applicable detailed specification.

4.3.8.4 Dimensions and tolerances. Launcher dimensions shall be minimum consistent with functional requirements and physical interface requirements. Unless otherwise specified, dimension tolerances shall be in accordance with DoD-D-1000.

4.3.8.5 Clearances. Launcher clearances shall be such that deflections of launcher elements during operational usage shall not adversely affect launcher, missile or aircraft performance and safety.

4.3.8.6 Launcher installation alignment. The launcher design shall provide for a combined missile/launcher installation which maintains missile alignment tolerances within those specified for prelaunch alignment of the missile and aircraft reference axes.



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4.3.8.7 Umbilical alignment. The launcher shall provide for alignment of the launcher to missile umbilical connector when the missile engages the launcher such that proper mechanical and electrical mating is achieved. In the case of a fixed installation, the same requirement applies to the launcher to aircraft umbilical connector. In the case of a jettisonable installation, the same requirement applies to the launcher-to-ejection rack umbilical interface.

4.3.8.8 Jettison. Jettison shall be required of the launcher and launcher/missile combination, but only when the launcher is installed on bomb racks. There shall be no circuitry for separate missile-only jettison.

4.3.8.9 Service access. The launcher design shall provide adequate access for servicing of the launcher and removal of foreign deposits. The design shall permit removal of all launcher power supplies and electronic assemblies and harnesses for repair.

4.3.8.10 Test access. The launcher design shall provide access and provisions for test of the launcher and retain compatibility with paragraph 5.12.

4.3.8.11 Drain provisions. Drain holes shall be provided as required to prevent collection and entrapment of water or other fluids. Where it is impractical to provide drain holes, cavities shall be sealed to prevent fluid penetration.

4.3.9 Human engineering. In addition to complying with MIL-STD-1472 and MIL-H-46855, the following additional requirements shall be incorporated in the design:

a. The anthropometric percentile range for armament ground crew shall be as specified by the procuring activity.

b. An unloaded launcher shall be capable of being handled, serviced and transported outside of its aircraft installation without any requirement for ancillary equipment (ie, carriers, bolts, crates, etc).

c. Any positional indication and instructions located externally shall be flush with the general outside surface.

d. All normal launcher functional operations shall be performed without the use of special tools.

e. All normal functional operations and indications (ie, latching, loading, locking, stray voltage checks, etc) shall be available from both sides of the launcher.

f. No components shall be accessible externally which would allow accidental or inadvertent firing or operation of the launcher.

g. All operation of the launcher shall require no more than two persons at any one time.



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h. Criteria shall be applied to the design to (1) assure that the equipment can be efficiently, safely, and reliably maintained and operated; (2) assure that adequate handling provisions have been included; (3) minimize human error type failures; (4) assure that design features will not constitute a hazard to personnel.

i. Early design effort shall include identification of human factor variables which are most likely to require detailed study for research during later design stages: ie, human performance requirements which may exceed human capabilities, degrade system objective, reflect possible unsafe practices, or may be prone to human error.

j. Access with sufficient internal space shall be provided for servicing, adjusting, etc.

k. The design of the launcher shall be such that there are no protrusions or critically located components that could easily be damaged by normal handling or normal operation of the launcher outside of a shipping container.

4.3.10 Environmental and service conditions. The launcher shall function properly under any and all combinations of environment experienced during launcher storage, ground operation and captive flight. The conditions specified herein shall be in accordance with the climatic criteria of MIL-STD-810 and MIL-T-7743 as modified by the operational service environments of ground operation and captive flight, and shall include:

a. Altitude: From sea level to 70,000 feet

b. Temperature: Captive flight mission profiles shall be developed to establish maximum time-temperature profiles. Low temperature conditions shall include worst case storage conditions and captive flight loiter cases, if applicable.

c. Humidity: Under all conditions of service and storage.

d. Dynamic Stresses: Under all conditions of vibration and shock that are present in a loaded and unloaded launcher during aircraft and launcher service operations.

e. Static Stresses: Under all conditions of high-g and static loads that are present in a loaded and unloaded launcher during aircraft and launcher service operations.

f. Contaminants: Under all conditions of service or storage.

4.3.11 Maintainability. Cleaning requirements shall be minimized, and cleaning shall be accomplished without removal of the launcher from the aircraft. The launcher shall be designed with component accessibility which requires a minimum of maintenance effort and facilities at all maintenance levels. The design shall provide for ease of assembly and disassembly with a minimal need for removal of hardware and special tools, test facilities and other support equipment for servicing. The use of modular assemblies shall be maximized for ease of replacement with minimum requirements for special skills. The shop-replaceable assembly (SRA) concept shall be employed.



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4.3.11.1 Maximum maintenance time. Design shall provide for:

a. With launcher removed from the aircraft a mean time to repair (MTTR) the launcher of one hour and a two hour maximum to include fault isolation to the component level, replacement of the failed component and checkout.

b. The launcher shall be designed to require a minimum number of maintenance hours.

4.3.11.2 Maintenance plan. The launcher shall be designed to be maintained at only the organizational and intermediate levels of maintenance in accordance with the policy, requirements, and procedures established in OPNAVINST 4790.2 of the Naval Maintenance Program. These two levels provide for a separation of the various maintenance tasks according to complexity, depth, facility, equipment requirements and personnel skills. The launcher shall be maintained at activities both ashore and afloat.

4.3.11.2.1 Organizational maintenance. Maintenance tasks performed at this level will be confined primarily to servicing and checkout of the launcher and shall be performed on a scheduled basis.

4.3.11.2.1.1 Organizational maintenance tasks. Maintenance at this level shall consist of the following:

a. Preflight and postflight operations such as visual inspection, cleaning, servicing and checkout.

b. Troubleshooting and testing conducted with launcher mounted to aircraft.

c. Removal and replacement of designated components without requirement for alignment and adjustment.

4.3.11.2.2 Intermediate maintenance. Maintenance tasks performed at this level shall be confined to support of organizational activities.

4.3.11.2.2.1 Intermediate maintenance tasks. The requirement for intermediate maintenance and inspection shall be minimized. Intermediate maintenance tasks shall consist of the following:

a. Inspection: Visual and dimensional inspections to determine corrosion damage and other defects reported by the organizational maintenance activity that would prevent reuse.

b. Repair by adjustment, servicing, and replacement of components, assemblies, and subassemblies not within the capabilities of organizational maintenance.

c. Checkout: Electrical and functional.



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4.3.11.2.3 Depot maintenance. Normal repair shall be limited to intermediate level. Only unscheduled items beyond capability of maintenance at the intermediate level will be repaired at the depot level. Depot level maintenance consists of repair, rebuilding of parts, assemblies, subassemblies, and end items, and condemnation.

4.3.11.2.3.1 Depot maintenance tasks. No scheduled depot maintenance actions are required.

4.4 Interchangeability. All parts having the same part number shall be in accordance with MIL-I-8500. All parts having the same part number shall be functionally and dimensionally interchangeable regardless of manufacturer. The item identification, manufacturer's part number, and part number requirements of DoD-D-1000 shall govern the part numbers and changes thereto.

4.5 Identification and marking. Identification marking shall be in accordance with MIL-STD-130. Nameplates for equipment identification shall be in accordance with MIL-P-15024, Size 3, Type A (c) or H, Style III and MIL-N-18307, and shall have space available for future marking of technical directives. All parts shall be marked with part numbers and, if applicable, lot numbers and serial numbers. Specifications on engineering drawings shall provide for part marking and shall indicate location of marking for accessibility.

4.6 Recycled, virgin and reclaimed materials. There is no exclusion to the use of recycled or reclaimed materials and no mandate for the use of virgin materials as long as it meets the requirements of this standard.



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## 5. DETAILED REQUIREMENTS

5.1 Loading of missiles. The launcher/missile interface shall be designed with adequate space provisions to allow safe, rapid loading and off-loading of all operationally configured missiles with a minimum of personnel, no peculiar equipment or tools required. The distance from the bottom of the launcher to the top of the missile as well as relative launcher/missile deformations shall conform to MIL-A-8591. This interface shall provide good visibility and easy access of all controls, connections, switches and devices that relate to missile loading/off-loading operations and positive visual determination of status of missile installation. Consideration shall be given to providing release openings for the use of hoisting equipment in the double-reeved configuration. If the hoist is attached to the launcher, the standoff capability of the hoist shall be considered. The launcher structure shall be sufficiently strong to react loads from the trolleys. If a missile uses hoisting equipment, single-hoisting-point lift (single-reeved) or double-reeved configurations shall be used with suitable missile attaching hardware. Consideration shall be given to missile handling equipment installation (including manual loading equipment) and hook-hanger entry angle during loading/off-loading operations. Launcher geometry shall provide for the proper application of existing non-permanent hoisting adapters, hoisting devices including manual loading equipment, and loading carts or trucks as specified herein. All loading/off-loading or related equipment attach or installation points shall be clearly marked and identified as to proper usage. Launcher/missile interface shall be compatible with the controls, load application range, and operational requirements of current U.S. Navy inventory support equipment (SE). In addition, the interface shall provide compatibility with manual loading equipment such that missiles can be loaded rapidly and safely with a minimum of personnel and without peculiar tools or equipment. All connections/disconnections, ie, arming and electrical connections, motor ignition circuit connections and safety interlock functions required during this sequence of events shall be accomplished after the missile is structurally attached to the launcher without releasing the missile attachment even if the missile is inadvertently fired. Alignment and contact between essential mating elements of the missile and launcher shall automatically accommodate a missile mispositioning of ±1 inch horizontally, ±1.0° in roll attitude and ±4° in pitch attitude or any combination thereof and accomplish required interfacing. The time required to complete all the steps necessary to reload the launcher shall be kept to a minimum with a design objective of not more than 5 minutes. The launcher shall provide for proper application of existing non-permanent hoisting adapters, hoisting devices including manual loading equipment, and loading carts or trucks, as specified in the following documents:

1. NAVAIR 19-15BD-6
2. NAVAIR 19-100-2
3. NAVAIR 19-100-1.1 (Volume 1) and 19-100-1.2 (Volume 2)



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All loading and off-loading or related equipment attachment or installation points must be clearly marked and identified as to proper usage. For Navy aircraft applications, any launcher loading attachments or hoisting components shall be capable of withstanding a vertical limit loading of 2.67 g and carrier roll rates of ±20 degrees with a 17-second period and pitch of ±3 degrees with an 8-second period. The launcher/missile interface shall be compatible with the controls, load application range, and operational requirements of current applicable U.S. inventory support equipment including the Single Hoist Ordnance Loading System (SHOLS).

5.1.1 Support Equipment. The launcher shall be designed so as to be capable of being installed, tested, serviced and maintained using standard support equipment (SE) and hand tools. Use of common hand tools shall be maximized. Special SE may be identified for the above functions only when required and approved by NAVAIR. All standard and special SE shall be identified and grouped by maintenance level (Organizational and Intermediate) according to the recommended maintenance actions.

5.2 Launcher structure. The launcher shall be capable of withstanding missile captive flight, launch and jettison loads and moments and shall transmit these from missile attach points to the aircraft through the shortest possible load paths. The launcher shall provide restraint to the captive missile against aerodynamic and inertia loads produced in the vertical, lateral and longitudinal axes and shall, if required, incorporate simplified anti-sway devices to prevent relative motion between the missile and the launcher. The structure shall be rigid enough to maintain the alignment requirements of 5.5.

5.3 Launcher to aircraft interfaces. When the launcher is hard-mounted to the aircraft structure, the launcher support bolt pattern shall conform to that of the LAU-7 launcher unit, and the electrical connectors shall be in compliance with MIL-STD-1760. Design loading requirements shall be in accordance with MIL-A-8591. For all rack mounted launchers, the physical interfaces shall comply with MIL-STD-2088, design loading requirements shall be in accordance with MIL-A-8591, and the electrical interfaces shall comply with MIL-STD-1760.

5.4 Launcher-to-missile physical interfaces. The launcher-to-missile electrical interfaces shall comply with MIL-STD-1760. Design loading requirements shall comply with MIL-A-8591.

5.5 Launcher interface alignment. The launcher shall provide boresight alignment of the missile consistent with aircraft/missile guidance requirements and as specified in the detailed specification. As a guideline, the misalignment of the missile and aircraft reference lines for each of the pitch, roll and yaw axes shall not exceed ±0.002 radian maximum ($3\sigma$).

5.6 Suspension and release system. The launcher suspension and release system shall provide positive, locked retention of the missile and shall release the missile only upon deliberate launch command or by proper manual release procedures. The launcher suspension system shall be designed to mate with available missile support hooks/hangers and to satisfy installation geometry requirements imposed by the aircraft weapon station.



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5.6.1 Suspension rail. The basic launcher structure shall consist of a rail beam that engages the missile hooks/hangers and allows for free longitudinal motion of the missile and restrains it against all other forces and moments.

5.6.2 Restraint system. The restraint system shall consist of a detent assembly which prevents longitudinal motion of the missile. The detent assembly shall include an electromechanical locking device to prevent motion of the detent latch and inadvertent release of the missile. Missile motion occurs when the locking device is removed, permitting detent latch override when the missile motor thrust reaches a prescribed level. Upon command from the aircraft, the interlock system shall be capable of unlocking the detent latch and completing the motor firing circuit. In the event of an abortive missile release, the interlock system shall provide automatic relocking of the detent latch and interruption of the motor firing circuit. In addition, if the motor inadvertently fires, the launcher may fail or be damaged, but shall retain the missile. The probability of inadvertent motor fire signal while in a locked condition shall be less than $1 \times 10^{-6}$. Also, the launcher restraint system shall be designed for crash safety conditions. During a normal launch, the detent latch mechanism shall be designed to return to its locked position after the missile has left the launcher. The restraint system shall be designed to permit quick and easy loading and downloading of the missile and to facilitate on-aircraft serviceability checks during ground operation.

5.6.3 Missile snubbing. Snubbers are used to eliminate all sway of the missile in the launcher during captive flight. Snubbers shall be designed so that this rigidity is removed by first missile motion during launch to allow free motion of the missile along the rail. Snubbers shall be easily releaseable during downloading.

5.7 Umbilical system. The launcher shall have provisions for an integral electrical umbilical system which mates with the missile and aircraft electrical connectors and transmits electrical signals to and from the aircraft and missile. Mating of the missile/launcher electrical connectors shall be manually or automatically accomplished, but either shall be possible after and while the missile is latched to the launcher. Connectors are to comply with applicable portions of 4.2.1.9. The launcher half of the electrical umbilical at the launcher/missile interface shall be protected so that when the launcher is not loaded with a missile, foreign substance contamination and possible electric shock to personnel are prevented.

5.8 Signal management system. The signal management system shall be capable of modifying all signals from external sources to supply the type and quantity required by the applicable missiles during ground operation and checkout, captive flight, and launching sequence. The system shall have appropriate short circuit and overload protection, and shall meet performance requirements under all applicable environments.

5.9 Fuzing and arming control. The launcher shall contain provision for mechanically or electrically initiating the missile's electromechanical arming system for arming the missile motor and missile warhead.



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5.9.1 Warhead arming provisions. If required, the launcher design shall transmit to the missile any electrical or mechanical signals necessary to initiate the missile's peculiar fuze arming sequence.

5.10 Missile sensing. The launcher shall provide for a missile present/gone indication capability. The sensing device shall have the capability for remote missile status indication. Missile sensing shall assure that hang fire or a hung missile retained by partial support provisioning is sensed as a missile present. The sensing device shall provide, by logic signal to aircraft control system or otherwise, the capability to deactivate all release and control signals to the launcher when no missile is present, except it shall not affect mechanical operations. Included shall be the capability for release and control system check out and testing other circuitry present without directly overriding the sensing device. The launcher sensing device shall be protected from damage at all times. Normally a missile launcher accepts missile present/gone signals through the missile umbilical connector and transmits the signal to the aircraft through the aircraft interface connector.

5.11 Safety in design. For adequate launcher design an active safety program shall be implemented and specific requirements introduced into the launcher design and development.

5.11.1 Launcher safety program. Launcher safety shall be achieved through pre-engineering safety analyses and evaluation of design concepts. This effort is required to provide surveillance, control and visibility to insure that no design creates a condition that can cause personnel injury.

5.11.1.1 Safety program functions. The safety program shall contain the following MIL-STD-882 requirements, as a minimum:

a. Provide for qualitative systems analyses to identify potential hazards. Close attention shall be given to those design features and characteristics that have created safety related problems on previous systems.

b. Submit periodic safety program status reports identifying all hazard levels, including probability of occurrence. Probabilities may be based on MTBF data and past failure histories. If a corrective safety program has been identified and is to be corrected within the scope of the contract, the corrective action also shall be included in these reports.

c. Safety program personnel shall participate in all design reviews as part of the overall safety effort.

d. The safety program objectives are:

(1) Eliminate all catastrophic and critical hazard levels and provide control of all marginal hazard levels commensurate with mission objectives. To select an optional corrective action and to determine whether to implement the corrective action, a tradeoff study shall be conducted. Final implementation shall be acceptable to the procuring activity.



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(2) Insure the subassemblies or components cannot be misassembled or misinstalled.

5.11.1.2 Failure mode effects analysis. During the launcher development process, the launcher assembly shall be analyzed for possible failure modes resulting from anticipated environmental, handling, and load conditions. This analysis shall evaluate the likelihood of occurrence of identified failure modes. For each failure mode identified, a determination shall be made of the effect of the part, circuit, or unit in question and of the ultimate significance of this effect relative to overall system safety.

This analysis shall include a discussion of the factors inherent in the design or the quality program that will minimize the likelihood of occurrence of those failures having the most significant effect on system safety. Periodically throughout the launcher development process, this analysis shall be updated as design configuration and load conditions change. Periodically throughout the development process, this analysis shall be reviewed for approval.

5.11.1.3 Specific design safety. The following specific safety features shall be considered in design:

5.11.1.3.1 Safety interlock. A safety interlock shall be furnished on the launcher which meets the requirements of 5.6.2 and 5.12.

5.11.1.3.2 Inadvertent release. To the maximum extent possible, inadvertent release or arming of a missile shall not occur due to the failure or malfunction of one single component of the launcher, and inadvertent motor firing shall not result in missile release.

5.12 Electromagnetic environmental effects ($E^3$). The requirement for control of electromagnetic environmental effects shall be as specified in 5.12.1 through 5.12.9.

5.12.1 Electromagnetic compatibility program (EMCP). A system EMCP in accordance with the requirements of MIL-E-6051 shall be applied to the launcher system and launcher system/aircraft, SE and GFE interfaces. This program shall be supported by an Electromagnetic Compatibility Advisory Board (EMCAB) established in accordance with the requirements of AR-43 for compliance with MIL-E-6051. The specifics for the Electromagnetic Compatibility Program Effort are provided in the Electromagnetic Compatibility Program Plan.

5.12.2 Electromagnetic compatibility (EMC). The launcher electrical system shall meet the requirements of MIL-E-6051 and MIL-STD-461 Class A1b using the test methods described in MIL-STD-462 (Notice 2). The following MIL-STD-461 tests are applicable: CE01, CE03, CE07, CS01, CS02, CS06, RE01, RE02, RS01, RS02, and RS03 or in accordance with the EMCP.

5.12.3 Electromagnetic vulnerability (EMV). The launcher electrical system shall meet the EMV requirements of MIL-E-6051 when subjected to the EMV environment of MIL-STD-461, as modified by table I and where applicable, MIL-HDBK-235.



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5.12.4 Lightning protection. The launcher system shall meet the requirements of the EMCP to provide compliance, where applicable, with the requirements of MIL-E-6051 and MIL-B-5087.

5.12.5 Bonding and grounding. The launcher system shall meet the requirements of the EMCP to provide compliance, where applicable, with the requirements of MIL-E-6051 and MIL-B-5087.

5.12.6 Wiring and cabling. The launcher system shall meet the requirements of the EMCP to provide compliance, where applicable, with the requirements of MIL-E-6051.

5.12.7 Static electricity. The launcher system shall meet the requirements of the EMCP to provide compliance, where applicable, with the requirements of MIL-E-6051.

5.12.8 Hazards of electromagnetic radiation to ordnance (HERO). The launcher system shall meet the HERO requirements of MIL-STD-1385 and MIL-E-6051.

5.12.9 Coupling factor (CF). The launcher shall be considered a potential coupling device which may provide entry for excessive radio frequency energy from the external environment, through the launcher into the remainder of the weapon system and possibly the entire aircraft system. The launcher shall utilize sufficient shielding, filtering or such other electromagnetic interference suppression techniques as may be required to ensure that the external electromagnetic environment shall not be coupled through the launcher. The external electromagnetic environment is as defined in MIL-STD-461 as modified by table I and, where applicable, MIL-HDBK-235. The radio frequency levels conducted from the launcher shall not be greater than those cited in MIL-STD-461, CE03. The CE03 limits shall be extended from 50 MHz to 100 MHz with no change and relaxed at 6 decibels (db)/octave above 100 megahertz (MHz).



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TABLE I. External electromagnetic environment (EME).

| Frequency (MHZ) | Field intensity | | PRF (pulses per sec) | Modulation pulse width (microsecond) |
|---|---|---|---|---|
| | VRMS/M 1/ | VRMS/M 2/ | | |
| 0.014-1.5 | | 10 | CW | ------- |
| 1.5-30 | | 200 | * | ------- |
| 30-200 | | 200 | CW | ------- |
| 200-225 | | 200 | 180-200 | 4-200 |
| 225-400 | | 100 | CW | ------- |
| 400-450 | | 200 | 300 | 0.6-60 |
| 450-840 | | 200 | CW | ------- |
| 840-970 | | 200 | 450-500 | 1.2-1.3 |
| 970-1215 | | 200 | CW/100-800 | 0.3-1.5 |
| 1215-1365 | | 200 | 140-3000 | 1-5 |
| 1365-2700 | | 200 | CW | ------- |
| 2700-3700 | | 200 | 200-3000 | 0.1-40 |
| 3700-5400 | | 200 | 10000 | 0.1 |
| 5400-5900 | | 200 | 425-500 | 0.3-3 |
| 5900-8500 | | 200 | CW/1-10,000 | 0.01-26 |
| 8500-10000 | | 200 | 200-2000 | 0.2-10 |
| 10000-10680 | | 200 | CW/1-10,000 | 0.01-26 |
| 10680-13200 | | 20 | CW | ------- |
| 13200-14400 | | 200 | 60,000 | 0.1-0.4 |
| 14400-15400 | | 20 | CW | ------- |



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TABLE I. External electromagnetic environment (EME) - Continued.

| Frequency (MHZ) | Field intensity VRMS/M 1/ | Field intensity VRMS/M 2/ | PRF (pulses per sec) | Modulation pulse width (microsecond) |
|---|---|---|---|---|
| 15400-16300 | | 200 | 3000 | 0.3 |
| 16300-16600 | | 200 | 1500 | 0.5 |
| 33000-33400 | | 200 | 2000 | 0.2 |

1/ Navy EME Test Levels, see EMCP or MIL-HDBK-235 if applicable.
2/ Minimum launcher EME specification requirements.

Legend:
CW - Continuous Wave
PRF - Pulse-Repetition Frequency
VRMS/M - Volts, Root Mean Square per Meter

* On/off keying Al modulation must be used at a 25 percent duty cycle.



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5.13 International standardization agreements. Certain provisions of this document are the subject of international standardization agreements. When change notice, revision, or cancellation of the document is proposed which will affect or violate the international agreement concerned, the preparing activity shall take appropriate reconciliation action through international standardization channels, including departmental standardization offices, if required. Table II identifies the paragraphs containing the requirements for the international standardization agreements.

TABLE II. Paragraghs containing international standardization agreements.

| Paragraph | Paragraph title | International standardization |
|---|---|---|
| 4.3.10 | Environmental and service conditions | AIR STD 20/18 |
| 5.12.8 | Hazards of electromagnetic radiation to ordnance (HERO) | AIR STD 20/16 |

Custodians
Navy - AS

Preparing Activity:
Navy - AS
Project No. 1440-N065

Applicable International Organizations:
North Atlantic Treaty Organization (NATO)
Air Standardization Coordinating Committee (ASCC)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions -- Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| | |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify:* _____ |

**5. PROBLEM AREAS**

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

**6. REMARKS**

| 7a. NAME OF SUBMITTER *(Last, First, MI)* — Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* — Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* — Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |
| | |

DD FORM 1426 PREVIOUS EDITION IS OBSOLETE.

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